

The pair singular series of a polynomial II: zeros of Dedekind zeta functions in the second moment of prime values

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Abstract

[draft; numbers to be filled from WP0–WP3] Let $f \in \mathbb{Z}[t]$ be irreducible without fixed prime divisor, $K = \mathbb{Q}[t]/(f)$, and $S_f(h)$ its pair singular series. In part I we proved that the diagonal part of $\sum_{h \leq H} (S_f(h) - C(f)^2)$ has Dirichlet series $D_f(s) = \zeta_K(s+1)E_f(s)$. Here we determine the analytic structure of E_f completely: it is an infinite product of Artin L -functions of virtual representations of the Galois group, the first of which is $L(2s+2, \text{Sym}^2 V_f)^{-1} = (\zeta_K(2s+2) \prod_O \zeta_{K_O}(2s+2))^{-1}$ with K_O the fields cut out by the unordered pairs of roots; D_f is meromorphic in $\Re s > -1$, has real poles at $s = -1 + 1/k$, and $\Re s = -1$ is a natural boundary. Consequently the Riesz means of the diagonal satisfy an explicit formula, unconditional for $m \geq 2$, whose oscillating part is indexed by the zeros of these Dedekind zeta functions, with terms $x^{m-1+\rho/2}$, and whose smooth part contains $x^{m-2/3}$. For $f = t^2 + 1$ every coefficient is computed exactly and the formula is verified against the exact Riesz means to $x = 4 \cdot 10^7$: [WP0 result]. We prove $\Omega_{\pm}(x^{m-3/4})$ unconditionally, $O(x^{m-7/8+\varepsilon})$ after the real terms under GRH for ζ_K , and give the function-field version, where the formula is an exact finite sum. Assuming the Hardy–Littlewood pair conjecture with a power-saving error, these results describe the second-order term of the variance of the number of prime values of f ; we explain precisely why the zero-induced fluctuations lie below the scale at which that transfer is valid, so that no statement about prime values themselves and the zeros of ζ_K is claimed.

1 Introduction

Recollection of part I. [copy the definitions: $\omega_f, C(f), S_f, b(q), W_f, a_f(d) = W_f(d)\omega_f(d), A_f = a_f * 1$, Theorems 2–4 of part I.] The identity

$$\sum_{h \leq H} (S_f(h) - C^2) = -C^2 \sum_{d \geq 2} a_f(d) \left\{ \frac{H}{d} \right\} + C^2 \text{Off}_f(H)$$

splits the sum into a diagonal, generated by $D_f(s) = \sum_d a_f(d)d^{-s} = \zeta_K(s+1)E_f(s)$ with $\text{Res}_{s=0} D_f = 1/C(f)$, and an off-diagonal over pairs of distinct roots. Conjecture 1 of part I, $\sum_{h \leq H} (S_f(h) - C^2) = -\frac{1}{2}C \log H + A_f + o(1)$, is equivalent (Cesàro form) to $\text{Off}_f^*(H) = o(H \log H)$.

The linear case as template. For $f(t) = t$, Goldston and Suriajaya [6] proved that the Riesz means $\sum_{k \leq x} (x-k)^m \mathfrak{S}(k)$, $m \geq 2$, equal $x^{m+1}/(m+1) - \frac{1}{2}x^m(\log x - H_m + \gamma + \log 2\pi) + x^{m-1} \sum_{|\gamma| \leq U} a(\rho)x^{\rho/2} + O(x^{m-1+\varepsilon})$, the zeros of ζ entering through the factor $1/\zeta(2s+2)$ of the generating series; and $\Omega_{\pm}(x^{m-3/4})$. [cite also Vaughan 2001, Friedlander–Goldston 1995, GS JNT 2021.]

Results. [one paragraph per theorem, in the order of the sections.]

What is conditional on what. [Reuse part I's paragraph (A)/(B)/(C) and add:] (D) Everything in Sections 2–4 is unconditional and concerns the arithmetic function A_f ; the Riemann hypothesis for ζ_K enters only in the sharp form of the error term. (E) Section 7 is conditional on the Hardy–Littlewood pair conjecture for f with an error term $O(T^{1/2+\varepsilon})$, as in Montgomery–Soundararajan [7]; the zero terms of the explicit formula are of relative size $T^{-1/4}$, *below* that tolerance. Unconditionally, the low-lying zeros of ζ_K govern the second-order fluctuations of the Riesz means of S_f ; under a Bateman–Horn pair conjecture with power saving these means control the variance of prime values of f only at a scale above the zero-induced fluctuations; and under GRH the variance is in turn equivalent, by the Goldston–Montgomery theorem [5] in the form of Bui, Keating and Smith [2], to pair correlation of the zeros of ζ_K . We claim no unconditional link between prime values of f and zeros of ζ_K .

Relation to the literature. [from LITERATURE.md §0: Kuperberg–Rodgers–Roditty–Gershon (prime elements of K , $\zeta_K(s+1)$ in a singular-series average, no explicit formula); Bhowmik–Halupczok–Matsumoto–Suzuki and Goldston–Suriyaya 2023 (multi- L explicit formulas in the Goldbach setting); Hall–Keating–Roditty–Gershon and Hochfilzer (variances governed by higher-degree L -functions over $\mathbb{F}_q[t]$); no prior appearance of $\text{Sym}^2 V_f$ or of a polynomial singular-series explicit formula.]

2 The analytic structure of D_f

2.1 The pair fields

Lemma 1 (pair fields). [PROVED] Let V_f be the permutation representation of $G = \text{Gal}(f)$ on the roots. Then $\text{Sym}^2 V_f \cong V_f \oplus V_f^{(2)}$, where $V_f^{(2)}$ is the permutation representation on unordered pairs of distinct roots. Consequently

$$L(s, \text{Sym}^2 V_f) = \zeta_K(s) \prod_O \zeta_{K_O}(s),$$

the product over the G -orbits O on unordered pairs, K_O the fixed field of the stabiliser of a pair in O . If G is 2-transitive there is one orbit and $K^{(2)} = \mathbb{Q}(\alpha + \beta, \alpha\beta)$ has degree $n(n-1)/2$.

Proof. $\text{Sym}^2 V_f$ has the basis $e_\alpha e_\beta$, $\alpha \leq \beta$, permuted by G ; the multisets $\{\alpha, \alpha\}$ span a copy of V_f and the remaining ones $V_f^{(2)}$. The L -function of a permutation representation $\text{Ind}_H^G \mathbf{1}$ is the Dedekind zeta function of the fixed field of H . $\square$

[Examples: quadratic: $\zeta^2 L(\chi_D)$; S_3 -cubic: ζ_K^2 ; S_4 -quartic: $\zeta_K \zeta_{K^{(2)}}$ with $[K^{(2)} : \mathbb{Q}] = 6$ (the resolvent sextic field); A_4 : same degree 6; D_4, C_4, V_4 : two or three orbits.]

2.2 The infinite product

Write $u = p^{-1}$, $v = p^{-(s+1)} = p^{-w}$. For p unramified in K the local factor of $D_f/\zeta_K(w)$ is

$$R_p(u, v) = \left(P_p + p b(p) \omega_f(p) p^{-s} \right) \det(1 - \text{Frob}_p v \mid V_f), \quad R_p(0, v) = (1 + \omega(g)v) \prod_i (1 - v^{\lambda_i(g)}),$$

$g = \text{Frob}_p$ of cycle type $\lambda(g)$ on the roots.

Theorem 2 (structure of E_f). [TO WRITE; proof is Möbius inversion + the computation of the script for the mixed terms] Let $c_k(g)$ be the coefficient of v^k in $\log R(0, v)$ (a class function on G with values in $\mathbb{Q}$), and define the virtual characters $\Psi_k = \frac{1}{k} \sum_{j|k} \mu(k/j) j c_j(\cdot^{k/j})$ [fix the exact inversion formula: it is the one solved level by level in *explicit-diag.py*; for G acting on two roots it gives the exponents of Table 1]. Then for every $K \geq 2$

$$E_f(s) = \prod_{k=2}^K L(k(s+1), \Psi_k) \cdot M_{f,K}(s),$$

where $M_{f,K}$ is an Euler product absolutely convergent for $\Re s > -1 + 1/(K+1)$ [check the exact abscissa: the pure- v remainder is $O(v^{K+1})$ and the mixed terms are $O(p^{-2}, p^{-2-w}, p^{-1-2w})$]. In particular D_f is meromorphic in $\Re s > -1$, with poles only at $s = 0$ and at the points where some $L(k(s+1), \Psi_k)$, $k \geq 2$, has a zero (if Ψ_k has a negative multiplicity there) or a pole (if positive), i.e. at $s = \rho/k - 1$ and, for the trivial constituents, at $s = -1 + 1/k$.

Table 1: $f = t^2 + 1$: exponents (a_k, b_k) of $\zeta(kw)^{a_k} L(kw, \chi_{-4})^{b_k}$ in E_f , $w = s + 1$ (exact; followup/scripts/explicit-diag.py). The real poles are at $s = -1 + 1/k$ for odd k ($a_k > 0$), of order a_k ; the zero families at $s = \rho/k - 1$ for $a_k < 0$ (ζ) or $b_k < 0$ (L).

k	2	3	4	5	6	7	8
a_k	-2	1	-1	3	-6	9	-14
b_k	-1	1	-2	3	-5	9	-16

Remark 3. [erratum to part I] Part I, Theorem 4, stated that the remaining factor $\tilde{H}_f$ after extracting $\zeta(3s+3)L(3s+3, \chi_D)$ converges absolutely for $\Re s > -1$. By Table 1 the pure- v expansion continues $(-3v^4 + 6v^5 - \dots)$, so $\tilde{H}_f$ converges absolutely only for $\Re s > -3/4$; its continuation to $\Re s > -1$ is the infinite product above. The conclusions of part I are unaffected.

2.3 The natural boundary

Theorem 4 (natural boundary). [PROVED modulo writing] D_f has zeros at every s with $p^{-s} = -P_p/(pb(p)\omega_f(p))$, $\omega_f(p) \geq 1$, and these accumulate at every point of the line $\Re s = -1$. Hence $\Re s = -1$ is a natural boundary of D_f , and of $\sum_n A_f(n)n^{-s} = \zeta(s)D_f(s)$. The same holds for $f(t) = t$, i.e. for the generating series of the twin-prime singular series.

Proof sketch. In $\Re s > -1 + 1/(K+1)$ write $D_f = \zeta_K(w) \prod_{k \leq K} L(kw, \Psi_k) M_{f,K}$; the absolutely convergent product $M_{f,K}$ vanishes exactly where one of its local factors does, and the local factor at p vanishes exactly where $F_p(s) = P_p + pb(p)\omega_f(p)p^{-s} = 0$ (the other factors are non-zero for $\Re w > 0$). The zeros of F_p have $\Re s = -\log(|P_p|/(pb(p)\omega_f(p)))/\log p \rightarrow -1$ and $\Im s = (2j+1)\pi/\log p$, which is dense in $\mathbb{R}$ as $p \rightarrow \infty$. A meromorphic continuation to a neighbourhood of a point $-1 + it$ would have that point as an accumulation point of zeros, hence a pole, near which there are no zeros: contradiction. [handle the finitely many ramified p and the poles of the L -factors; cite Estermann [4], Dahlquist [3], Bhowmik–Essouabri–Lichtin [1] for context.] $\square$

3 The explicit formula for the diagonal

Let $R_m(x) = \sum_{n \leq x} (x-n)^m A_f(n)$ and $G_m(s) = m! \zeta(s) D_f(s) / (s(s+1) \cdots (s+m))$, so that $R_m(x) = \frac{1}{2\pi i} \int_{(2)} G_m(s) x^{s+m} ds$.

Theorem 5 (quadratic case). [TO PROVE: Goldston–Suriajaya’s contour argument with Theorem 2; zero-density for ζ and $L(\chi_D)$] Let f be an irreducible quadratic of discriminant D , $m \geq 2$. For every $\delta > 0$,

$$R_m(x) = \frac{D_f(1)}{m+1} x^{m+1} + x^m \left(-\frac{\log x}{2C(f)} + c_f \right) + \sum_{\substack{k \text{ odd} \\ 3 \leq k < 1/\delta}} x^{m-1+1/k} Q_k(\log x) + x^{m-1} \sum_{\substack{k \geq 2 \\ k < 1/(2\delta)}} \sum_{\substack{\rho_k \\ |\gamma| \leq U_k}} x^{\rho_k/k} Q_{k, \rho_k}(\log x) + O(x^{m-2/3})$$

where ρ_k runs over the zeros of $\zeta(\cdot)^{-a_k} L(\cdot, \chi_D)^{-b_k}$ (those with negative exponent), Q are polynomials of degree less than the order of the pole, and $U_k \asymp x^5$. The $x^{m-2/3}$ term is present whenever $a_3 > 0$, i.e. for every quadratic.

Theorem 6 (general f). *[TO PROVE; statement as above with the Dedekind zeta functions of Lemma 1 at $k = 2$ and the $L(\cdot, \Psi_k)$ at $k \geq 3$. Hypothesis needed: the Ψ_k with negative constituents are (differences of) permutation characters, so that the relevant L -functions are quotients of Dedekind zeta functions and zero-density estimates are available. Check for S_n , A_n ; state the general case under Artin's conjecture + a zero-density hypothesis.]*

3.1 $f = t^2 + 1$: every coefficient

[Table of $a_m(\rho)$ for the first zeros of $L(\chi_{-4})$ and ζ ($m = 2, 3$), the real-pole coefficients, $D_f(1) = 0.7507908137$, $C(f) = 1.372813462821$, c_f . All from explicit-diag.py.]

3.2 Numerical verification

$f = t^2 + 1$, $m = 2$: data minus exact smooth terms (grey) and predicted zero sum (blue)

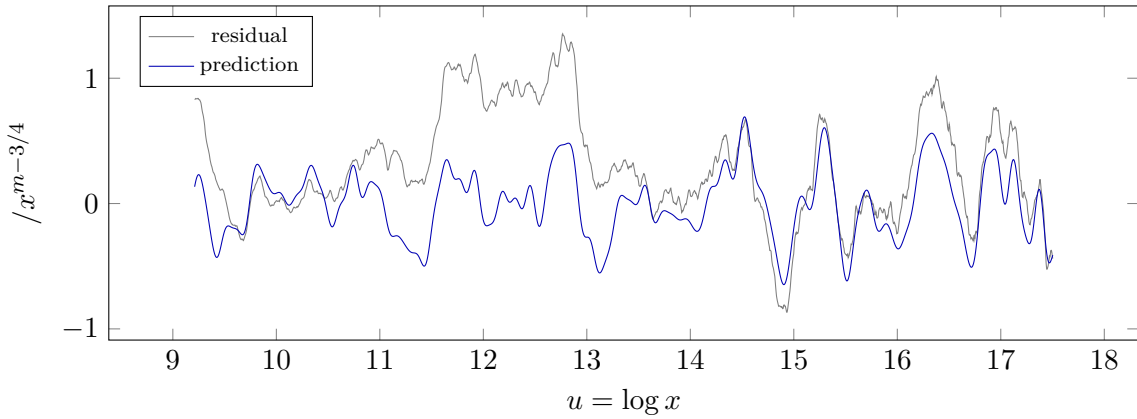


Figure 1: WP0. The residual of the diagonal Riesz mean of order 2 after subtracting the exact terms at $s = 1$, $s = 0$ and the real poles $s = -2/3, -4/5, -6/7$, against the sum over the zero families $k = 2, 4$ (zeros of $L(s, \chi_{-4})$ and of ζ with $\gamma \leq 100$). Regression coefficient $\beta = 1.008$ (0.991 with a smooth nuisance basis); per family $\beta_\zeta = 1.006$, $\beta_L = 1.020$.

[WP0: Figure of $(R_m - \text{smooth terms})/x^{m-3/4}$ against $u = \log x$ with the predicted zero sum overlaid; regression coefficient $\beta = 1.008$ ($m = 2$), 0.779 ($m = 3$); fraction of residual power explained; controls with shifted ordinates; per-family β ($k = 2$ zeros of L , of ζ ; $k = 4$). Discussion: what the range $10^4 \leq x \leq 4 \cdot 10^7$ can and cannot resolve (ordinates closer than $\approx 4\pi/8.3 \approx 1.5$ are not separated; this is a matched-filter test).]

4 Ω -results and conditional bounds

Theorem 7. *[TO PROVE: Landau's theorem as in GS Theorem 4; needs $A_f(n) \geq 0$, true when $P_p \geq 0$ for all p , e.g. $t^2 + 1$; otherwise use the first zero directly as in GS JNT 2021] For $f = t^2 + 1$ and every $m \geq 1$, $R_m(x) - (\text{smooth terms}) = \Omega_\pm(x^{m-3/4})$. Under RH for ζ and $L(s, \chi_{-4})$ and the simplicity of their zeros, the oscillating part is $O(x^{m-3/4+\varepsilon})$ and, for $m \geq 2$ [and a Gonek–Hejhal-type bound], an absolutely convergent sum $x^{m-3/4} \sum_\gamma a_m(\rho) x^{i\gamma/2}$, with remainder $O(x^{m-3/4-\varepsilon_0})$ for some $\varepsilon_0 > 0$. [Do NOT claim $O(x^{m-7/8+\varepsilon})$: the $k = 4$ family at $\Re s = -7/8$ has individually huge coefficients that cancel against the line integral ($|D_f|$ grows towards the natural boundary); numerically it is rejected by the data. Remark on asymptotic vs convergent explicit formulas near a natural boundary.]*

[Remark: the first zero of $L(\chi_{-163})$ at $\gamma = 0.2029$ (period 62 in $\log x$) for $t^2 + t + 41$: the "oscillation" is invisible in any computable range; class-number-one repulsion; Katz–Sarnak symplectic family statement for $L(\chi_D)$ (LITERATURE.md).]

5 The off-diagonal

[Route A of routes.tex: Prop. A1 (proved), A2/A3 (to prove) for quadratics. New here: the off-diagonal Dirichlet series $\sum_d d^{-s} W_f(d) \sum_{s \neq s'} \zeta(s, \langle (s' - s)/d \rangle)$ and the statement that Conjecture 1 of part I, in analytic form, is its continuation to a pole-free strip left of 0. Numerical: $\text{Off}_f(H)$ for $t^2 + 1$ to 10^7 and its spectrum. If nothing more: state clearly that the explicit formula of Section 3 is for the diagonal and that the full sum adds an $O(H)$ -in-Cesàro-form term whose fluctuations are those of the fractional parts of the root differences.]

6 The function field $\mathbb{F}_q[u]$

[Route B. New: with $T = q^{-s}$ the diagonal generating function is a rational function of T (Prop. B2 of routes.tex + Theorem 2), so the Riesz-mean analogue is an EXACT finite sum over the inverse roots: the Weil zeros of $L(T, \chi_D)$ and of the extracted factors, and the real "poles" $T = q^{1/k}$. Write it down for $f = t^2 - D$; compare with thesis/ff.ts data. Then Prop. B3, $q \rightarrow \infty$, for the off-diagonal (Katz/Deligne).]

7 Consequences for prime values

[Conditional section. (i) Under the HL pair conjecture with error $O(T^{1/2+\varepsilon})$, the variance formula of part I with the diagonal explicit formula inserted: the polynomial Montgomery–Soundararajan law plus an explicit oscillatory correction of relative size $T^{-1/4}$, which is below the error tolerated by the hypothesis — say so. (ii) Goldston–Montgomery for ζ_K : what Bui–Keating–Smith give for the degree-2 non-primitive $\zeta_K = \zeta L(\chi_D)$ (union of two independent GUE processes, Murty–Perelli), and why no analogue exists for prime values of a non-linear f (no Dirichlet series). (iii) What GRH for ζ_K does say as a theorem: Grenié–Molteni–Perelli for prime ideals; our $O(x^{m-7/8+\varepsilon})$.]

8 Discussion and open problems

[(1) Prove Theorem 6 in general. (2) The off-diagonal for quadratics (Route A). (3) Is the natural boundary known for the Goldston–Suriyaya series? (4) Higher moments: R_k for polynomial tuples and Gaussian fluctuations. (5) The constant A_f in closed form.]

Data and code

followup/scripts/riesz-raw.ts (raw Riesz means), followup/scripts/explicit-diag.py (exponents, constants, residues, comparison), followup/data/*. [add WP2 scripts]

Statement on authorship and the use of AI

[as in part I]

References

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